

The education-obesity link across countries over time

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Title Page

Tutorial group: 6.1

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Part 1 - Identify a Social Problem

The social problem we have identified is the link between the BMI Index and educational attainment level, bridging two data sets from Eurostat to compare and identify any correlation between the two factors.

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

The correlation between these two variables is relevant as it has real consequences for individuals, healthcare systems, and policymakers. The relevance of this lies in the fact that obesity comes with serious health risks, such as; diabetes and cardiovascular issues. This then results in the burden of health risks being disproportionately put on the lower educated class of society, and further exacerbating social inequality issues. Beyond the health risks themselves, education has a clear impact on the ability of a person to maintain a healthy lifestyle, as higher education often comes with higher income, better access to higher quality food and so on. This means that addressing obesity cannot be done through increasing awareness alone, and that broader structural interventions, such as food pricing policies or workplace conditions need to be considered.

The fact we are looking at the European aspect of this issue, also adds to its relevance. By looking at multiple countries over time, it becomes possible to assess whether this education gap in obesity rates is a universal pattern across Europe or whether certain countries have managed to narrow it. This kind of cross country comparison is useful for understanding what policy approaches may actually work in practice.

Part 2 - Data Sourcing

2.1 Load in the data

Both datasets are downloaded directly from Eurostat using the eurostat package, so the data is fetched when the document is knitted and nothing needs to be stored in the GitHub repository.

```
library(tidyverse)

# Load the saved datasets from our repository
bmi_edu <- read_csv("data/bmi_edu.csv")
obesity_trend <- read_csv("data/obesity_trend.csv")
```

2.2 Provide a short summary of the dataset(s)

```
head(bmi_edu)
```

```
## # A tibble: 6 x 9
##   freq unit bmi      isced11 sex age geo TIME_PERIOD values
##   <chr> <chr> <chr>      <chr> <chr> <chr> <chr>      <dbl> <dbl>
## 1 A PC BMI18P5-24 ED0-2 F TOTAL AT 2014 45.6
## 2 A PC BMI18P5-24 ED0-2 F TOTAL AT 2019 40.9
## 3 A PC BMI18P5-24 ED0-2 F TOTAL BE 2014 45.7
## 4 A PC BMI18P5-24 ED0-2 F TOTAL BE 2019 41
## 5 A PC BMI18P5-24 ED0-2 F TOTAL BG 2014 44.3
## 6 A PC BMI18P5-24 ED0-2 F TOTAL BG 2019 48.1
```

```
head(obesity_trend)
```

```
## # A tibble: 6 x 6
##   freq unit bmi      geo TIME_PERIOD values
##   <chr> <chr> <chr>      <chr>      <dbl> <dbl>
## 1 A PC BMI25-29 AL 2022 44.8
## 2 A PC BMI25-29 AT 2008 36.5
## 3 A PC BMI25-29 AT 2014 33.3
## 4 A PC BMI25-29 AT 2017 35
## 5 A PC BMI25-29 AT 2019 35
## 6 A PC BMI25-29 AT 2022 35
```

Our first dataset, `bmi_edu`, comes from Eurostat and is called “Body mass index (BMI) by sex, age and educational attainment level” (dataset code `hlth_ehis_bm1e`). It is based on the European Health Interview Survey (EHIS), a survey in which people across European countries report their own height and weight. For each country and year, the dataset gives the percentage of people in different BMI categories (such as normal weight, overweight, and obese), broken down by education level, sex, and age. Because the survey is only carried out every few years, the data covers a small number of waves (around 2014 and 2019). This dataset is central to our project because it is the one that connects weight to education level.

Our second dataset, `obesity_trend`, also comes from Eurostat and is called “Obesity rate by body mass index (BMI)” (dataset code `sdg_02_10`). It is one of Eurostat’s Sustainable Development Goal (SDG) monitoring indicators, built from European health surveys in which adults report their own height and weight. For each country and year, it gives the percentage of people in different BMI categories such as overweight and obese. Unlike our first dataset, it does not break the figures down by education level, but it covers more years per country (for example 2008, 2014, 2017, 2019 and 2022), which makes it well suited for showing how obesity has changed over time. This dataset is important for our project because it provides the “over time” and “across countries” side of our research question.

2.3 Describe the type of variables included

The variables in our datasets fall into a few types. The main health information is the BMI/obesity measure — the share of people who are overweight or obese (the `bmi` and `values` columns). The main socioeconomic (SES) information is the education level (the `isced11` column in `bmi_edu`), which is the variable at the centre of our research question. The remaining variables — sex, age, country (`geo`) and year (`TIME_PERIOD`) — are background characteristics that let us split the figures by group, country and time. In terms of how the data was collected, neither dataset comes from administrative sources such as hospital or government records. Both are based on surveys in which people are interviewed and report their own height and weight. This has two consequences. First, because height and weight are self-reported, obesity tends to be slightly underestimated, as people often round their weight down. Second, our two datasets are not fully independent, because the SDG indicator (`obesity_trend`) is partly built on the same European Health Interview Survey as `bmi_edu` — so they are related rather than two completely separate sources. We chose these two datasets together on purpose: `bmi_edu` provides the education breakdown we need, while `obesity_trend` gives a

longer view over time and across countries. Because both rely on self-reported survey data, our analysis can describe patterns and associations — for example, that obesity is more common among lower-educated groups — but it cannot prove that education causes obesity.

Part 3 - Quantifying

3.1 Data cleaning

Say we want to include only larger distances (above 2) in our dataset, we can filter for this.

Cleaning Step 1: Filter and clean the first dataset

```
bmi_edu_clean <- bmi_edu %>%  
  filter(  
    bmi == "BMI_OBS",           # Keep only Obese rows  
    isced11 %in% c("ED0-2", "ED3_4", "ED5-8"), # Separate the 3 education groups  
    sex == "T",                 # Combined sexes  
    age == "Y15-74"            # Combined adult age group  
  ) %>%  
  select(geo, TIME_PERIOD, isced11, bmi_value = values) %>%  
  drop_na(bmi_value)
```

Cleaning Step 2: Filter and clean the second dataset

```
obesity_trend_clean <- obesity_trend %>%  
  filter(  
    bmi == "BMI_OBS" # Keep only Obese rows since that's what we are comparing  
  ) %>%  
  select(geo, TIME_PERIOD, trend_value = values) %>%  
  drop_na(trend_value)
```

Cleaning Step 3: Combine both datasets into one master file

```
combined_dataset <- inner_join(bmi_edu_clean, obesity_trend_clean, by = c("geo", "TIME_PERIOD"))
```

Please use a separate 'R block' of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1

```
# Create Variable 1: Human-readable education levels
bmi_edu_with_var1 <- combined_dataset %>%
  mutate(
    edu_level = case_when(
      isced11 == "ED0-2" ~ "Low Education",
      isced11 == "ED3_4" ~ "Medium Education",
      isced11 == "ED5-8" ~ "High Education"
    )
  )
```

Variable 2

```
# Create Variable 2: Convert the percentage value to a 0-1 decimal proportion
final_analysis_data <- bmi_edu_with_var1 %>%
  mutate(
    bmi_proportion = bmi_value / 100
  )
```

3.3 Visualize temporal variation

3.4 Visualize spatial variation

Here you provide a description of why the plot above is relevant to your specific social problem.

3.5 Visualize sub-population variation

What is the poverty rate by state?

```
# {r visualise_subpopulations} dataset$inmetro <- dataset$inmetro %>% as.factor() #
Boxplot of poverty rate by state using the 'midwest' dataset ggplot(dataset, aes(x =
inmetro, y = percadultpoverty)) + geom_boxplot() + labs(title = "Distribution
of Poverty Rates by Metropolitan status (Midwest counties)", x = "Metropolitan Area",
y = "Poverty Rate of Adults (%)") + theme_minimal() + theme(legend.position =
"right")
```

Here you provide a description of why the plot above is relevant to your specific social problem.

3.6 Event analysis

Analyze the relationship between two variables.

Here you provide a description of why the plot above is relevant to your specific social problem.

Part 4 - Discussion

4.1 Discuss your findings

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: ...

5.2 Reference list

Use APA referencing throughout your document.

Eurostat. (2025). Overweight and obesity by sex, age and educational attainment level — SDG indicator [Data set]. European Commission. https://ec.europa.eu/eurostat/databrowser/view/sdg_02_10

Dataset 2 — European Health Interview Survey, BMI

Eurostat. (2023). Body mass index (BMI) by sex, age and educational attainment level — European Health Interview Survey [Data set]. European Commission. https://ec.europa.eu/eurostat/databrowser/view/hlth_ehis_bm1e